

# An Algebraic Framework for the Game of Go (Version 3)

## 0. Acknowledgment

This essay is a revised synthesis of earlier drafts. The author gratefully acknowledges detailed reviews that have dramatically tightened the framework presented here: the game tree is modeled as a history-dependent DAG to rigorously capture superko; liberties are derived via the coboundary operator; life and death are formalized via the Knaster-Tarski fixed point of Benson's operator; and endgames are analyzed through combinatorial game theory.

## 1. Introduction

The ancient board game of Go, played on a grid of  $n \times n$  intersections (typically  $n = 19$ ), has long been recognized as a fertile ground for mathematical inquiry. While much of the existing literature addresses Go from the standpoint of algorithmic complexity and statistical evaluation, this essay proposes a systematic algebraic formulation of the game. By casting Go in terms of posets, chain complexes, Boolean lattices, and the theory of surreal games, we demonstrate that the board and the dynamics of play admit natural mathematical structures. We prioritize rigorous abstraction: every algebraic object introduced here performs genuine work, yielding a mathematically exact translation of Go's combinatorial mechanics.

## 2. The State Space: Histories and the Superko DAG

Let  $B_n = \{(i, j) : 1 \leq i, j \leq n\}$  denote the intersections of an  $n \times n$  board. A static board position is a function  $p : B_n \rightarrow \{0, 1, 2\}$ , representing empty, black, and white intersections, respectively. Let  $\mathcal{P}_n$  denote the set of all such configurations.

Because the legality of a Go move depends on the Positional Superko rule (which forbids the exact repetition of any prior board state), legality is not a property of static boards, but of game *histories*. Define a **history** as a finite sequence  $h = (p_0, p_1, \dots, p_k)$  where  $p_0$  is the empty board and each  $p_i \in \mathcal{P}_n$ .

We construct the game state space as a directed graph  $\mathcal{H}_n$ :

- **Vertices** are valid game histories.
- **Edges** represent legal transitions: an edge exists from  $h = (p_0, \dots, p_k)$  to  $h' = (p_0, \dots, p_k, p_{k+1})$  if and only if  $p_{k+1}$  is a valid move from  $p_k$  and, crucially,  $p_{k+1} \notin \{p_0, \dots, p_k\}$ .

**Proposition 2.1.** *The graph  $\mathcal{H}_n$  is a finite directed acyclic graph (DAG). Therefore, its reachability relation forms a partial order. Proof.* Every edge in  $\mathcal{H}_n$  strictly increases the length of the history sequence. Because the superko rule forbids repetitions, the maximum length of any history is bounded by the cardinality of  $\mathcal{P}_n$  (which is  $3^{n^2}$ ). A directed graph where all paths are strictly length-increasing and bounded is a finite directed tree, which is trivially acyclic.  $\square$

This load-bearing finiteness ensures that every valid game terminates, providing the well-founded structure necessary for the fixed-point and combinatorial evaluations in later sections.

## 3. Topological Go: Coboundaries and Liberties

Model the board as a graph  $G = (V, E)$  where  $V = B_n$  and  $E$  comprises orthogonal adjacencies. Let  $C_0(G; \mathbb{Z}_2)$  and  $C_1(G; \mathbb{Z}_2)$  be the standard chain groups, with the coboundary operator  $\delta_0 : C^0 \rightarrow C^1$  defined by  $\langle \delta_0 \varphi, e \rangle = \varphi(u) + \varphi(v)$  for  $e = \{u, v\}$ .

A **stone group**  $c$  is a connected monochromatic component. Represent  $c$  by its indicator cochain  $\mathbf{1}_c \in C^0(G; \mathbb{Z}_2)$ . The cochain  $\delta_0(\mathbf{1}_c) \in C^1$  precisely identifies the edge cut separating  $c$  from its complement.

Let  $V_\emptyset$  denote the set of empty intersections. We define the **liberty set** of  $c$  algebraically:

$$L(c) = \{v \in V_\emptyset : \exists u \in c \text{ with } \{u, v\} \in \text{supp}(\delta_0(\mathbf{1}_c))\}.$$

A capture occurs exactly when  $\delta_0(\mathbf{1}_c)$  has no support on edges incident to  $V_\emptyset$ .

Furthermore, we define a **topological candidate eye** as a nontrivial cycle  $z$  in the first homology group  $H_1(G_c; \mathbb{Z}_2)$ , where  $G_c$  is the subgraph induced by  $c$ . Let  $\text{Int}(z)$  denote the set of empty vertices spatially enclosed by the cycle  $z$ . *Remark 3.1:* Topological enclosure ( $H_1$ ) is a necessary, but not sufficient, condition for a "true eye," as diagonal ko-fights (false eyes) depend on the dynamic history tree  $\mathcal{H}_n$ . Thus, homology identifies *candidate* enclosures, leaving ultimate verification to lattice theory.

## 4. Lattice Theory: The Fixed Point of Life and Death

Consider the Boolean lattice  $L = \mathcal{P}(\text{Black stones})$  ordered by inclusion. Define Benson's operator  $f : L \rightarrow L$ , where  $f(X)$  outputs the subset of stones in  $X$  whose liberties lie entirely within regions completely enclosed by  $X$ . Because  $L$  is a finite complete lattice and  $f$  is monotone, the **Knaster-Tarski theorem** applies:

**Theorem 4.1.** *The operator  $f$  possesses a greatest fixed point  $\text{gfp}(f) = \bigvee \{X \in L : f(X) \supseteq X\}$ . This set constitutes the unconditionally alive (pass-alive) Black stones.*

We can now rigorously bridge the topological candidate eyes of Section 3 with the lattice-theoretic vital regions of Section 4.

**Lemma 4.2 (Topological Bridge).** *Let  $c$  be a Black stone group. Suppose  $c$  possesses two disjoint  $H_1$ -cycles,  $z_1$  and  $z_2$ . If, for  $i \in \{1, 2\}$ ,  $\text{Int}(z_i)$  contains no opponent stones and  $\text{Int}(z_i) \subseteq L(c)$ , then  $c \subseteq \text{gfp}(f)$ . Proof.* The condition  $\text{Int}(z_i) \subseteq L(c)$  guarantees that every vertex inside the topological hole  $z_i$  is a liberty derived directly from the coboundary of  $c$ . The absence of opponent stones ensures these liberties cannot be reduced from within. Two such disjoint regions guarantee that no single opponent placement can capture  $c$ . Thus, the liberties of  $c$  are permanently secured within regions enclosed by  $c$ , satisfying the criteria for  $f$ -invariance. Therefore,  $c$  belongs to the greatest fixed point.  $\square$

## 5. Combinatorial Game Theory: Direct Sums and Temperature

As the board simplifies in the endgame, the global position often fractures into isolated skirmishes. Following Conway, Berlekamp, and Wolfe, each independent region  $R_i$  acts as a local game  $g_i$ . Because we established  $\mathcal{H}_n$  as a finite DAG, we can formally evaluate the global game as a **direct sum** in the group of surreal games:

$$G = g_1 \oplus g_2 \oplus \cdots \oplus g_k.$$

While strict regional independence is a strong assumption—real games feature global coupling via *ko* threats and *sente*—combinatorial game theory dynamically handles these interactions through the mathematics of **thermography**. The "temperature" of a local game  $g_i$  measures the urgency of playing there, and the operation of *chilling* allows the integration of locally coupled states into the broader global sum. Thus, evaluating the board reduces to computing the sum of these surreal values, bridging pure algebra with applied strategy.

## 6. Conclusion

This essay provides an algebraically rigorous framework for Go. By establishing the game history as a finite DAG, translating captures into coboundary operators, defining unconditional life via Knaster-Tarski fixed points, and evaluating endgames as direct sums of

surreal values, we demonstrate that Go's intuitive mechanics map flawlessly onto formal mathematics.